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Revisiting a Foot Sole Sensor Setups:

Reproducibility, Technical Challenges, and Future Validations

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PROJECT OVERVIEW

Background:

For most of human history, people walked barefoot. Shoes are a relatively recent invention, appearing roughly 40,000 years ago, and over time, they evolved into the cushioned, structured models we know today. For a long time, that development was seen as straightforward progress.

Yet in the late 2000s, with the rise of interest in minimalist and barefoot-style footwear, this assumption began to be questioned. Despite this, relatively little is known about a basic issue: Does the material of the sole actually make a difference? If the thickness is minimized to simulate the sensory and mechanical characteristics of barefoot walking, how would it differ from using a traditional shoe?

Previous contribution:

Project 2024 [1]	Project 2025 [2]
Wired Velostat sensor array inside Foot sole shoe	Wireless BLE communication using ESP32-C3
Calibration of Velostat sensors	BLE-enabled system with real-time UART bridge
Baseline Python software for data logging and visualization	Extended logging pipeline with BLE + database integration
Data transfer via USB-C serial connection	Wireless real-time BLE data transmission
Data stored only as .h5 files	Transformed data from HDF5 → CSV, and added an SQLite database for structured storage and queries
Python visualization pipeline	Reused and refined visualization pipeline with minor modifications
Output video only in .mp4 format	Added HTML-based interactive video generation
Run scripts manually to generate video	Added automated data processor interface for streamlined workflow
	Additional measurements: • Foot sole pressure sensitivity using monofilament • Sensitivity tests using tuning fork • Pressure sensitivity comparison using bare foot + sensor and foot sole + sensor
	Improved user experience with a 3D-printed enclosure for the BLE electronics

With the provided setup, we were fully motivated to find out:

How do thin leather soles (minimalist) and thin rubber soles (conventional) differ in their effects on foot sensitivity, plantar pressure distribution, and perceived barefoot-like sensation during standing and walking?

Hypothesis1: Perceptual Sensitivity

H1: Participants will report greater foot sensitivity (higher ratings of hardness, roughness, and stability) with the thin leather sole compared to the thin rubber sole, with leather being closer to true barefoot perception.

Rationale: Leather is stiffer and less damped than rubber, allowing better transmission of ground texture and vibration to plantar mechanoreceptors. Rubber may absorb high-frequency vibrations, reducing tactile feedback.

Hypothesis 2: Mechanical Pressure Differences

H2: The thin leather sole produce plantar pressure patterns closer to barefoot conditions (higher peak pressures, smaller contact areas, more dynamic center-of-pressure movement) compared to the thin rubber sole.

Rationale: Rubber soles deform more under load and have higher friction, potentially increasing contact area and dampening pressure gradients. Leather maintains more rigid pressure transmission similar to barefoot.

EXPERIMENTAL DESIGN

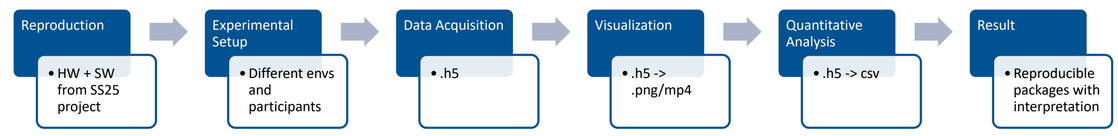
Three experimental conditions were evaluated to assess the influence of sole type on plantar pressure distribution and potential biomechanical implications:

1. Flat Surface: Participants walked 20 steps along a straight indoor corridor under controlled conditions**.**
2. Stair Ascent/Descent: Participants ascended approximately two floors (≈ 20 steps) and subsequently descended the same staircase**.**
3. Inclined Slope (≥ 5%): Participants walked 12 steps downhill and 12 steps uphill on a slope of at least 5% incline.

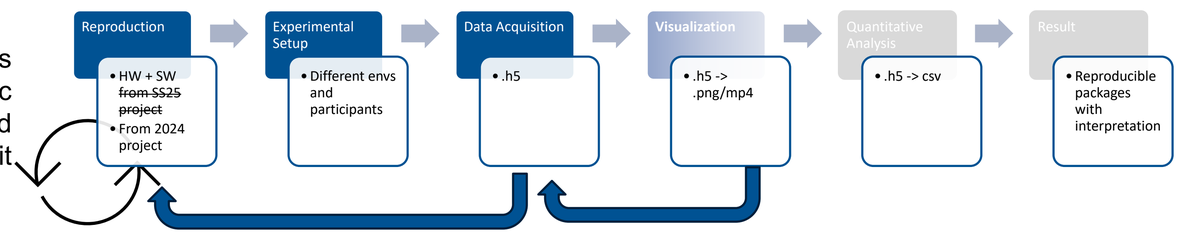
Each condition was performed using:
A flat leather shoe (Fullsoul)
The participant's regular rubber-soled shoe

Measurements were conducted separately for the left and right foot.

Planned Procedure:



Reality hits different:



PRELIMINARY VALIDATIONS

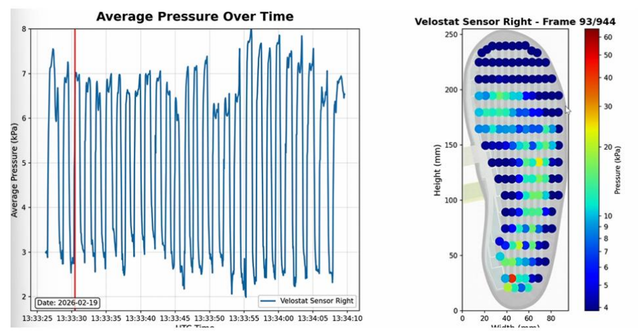
1. **Reproducibility:**

The original design relied on Bluetooth Low Energy (BLE) communication between the ESP32-based insole and the host computer. However, wireless transmission proved unstable and non-operational in repeated trials. In parallel, parts of the software pipeline (visualization and HDF5-to-CSV conversion scripts) required modification before execution.

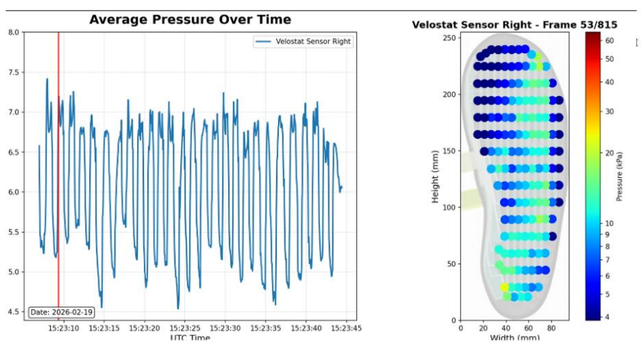
Hardware inspection further revealed structural fragility within the sensing assembly, including vulnerable conductive interfaces in the Velostat pressure matrix.

→ Given these constraints, the team implemented a **wired USB-based fallback solution**, replacing wireless transmission with direct cable communication.

2. **Preliminary Data Observations on Foot Sensitivity**



Leather sole



Conventional rubber sole

Based on visual inspection of the pressure matrices:
- The **leather sole (flat, minimal cushioning)** appears to produce more localized pressure concentrations, especially in the forefoot and central midfoot regions.
- The **rubber sole** appears to redistribute pressure more broadly, reducing localized peaks.

This suggests that the flat leather sole may allow more direct ground-force transmission, potentially increasing localized plantar loading, whereas the rubber sole provides partial shock absorption and pressure dispersion.

*However, these observations are preliminary and based on qualitative frame inspection. A full statistical analysis across multiple gait cycles and participants is required before drawing definitive biomechanical conclusions.

CONCLUSION

The research question remains open and requires further verification. With the lessons learned, improved documentation, and a clearer protocol, we believe the next group will be in a stronger position to validate whether the preliminary observations- including potential links to plantar fasciitis, over-pronation, and muscle atrophy- can be confirmed rigorously.

Our learnings through this journey:
Reproducibility ensures data integrity.
A system that cannot be reliably reproduced cannot generate scientifically defensible conclusions.
Backup strategies are essential.
Data loss and hardware failure can occur at any stage. Redundant logging, stable architectures, and fallback solutions are critical.
Communication accelerates problem-solving.
Early coordination with previous team members and transparent documentation significantly reduces troubleshooting time.

REFERENCES

[1] Burian, K., Ren, C., & Strauch, A. (2024, July 30). Foot sole pressure sensing device and its usage in barefoot shoes: Fullsoul Runningpad. Clinical Applications of Computational Medicine (CACOM), Technical University of Munich. Original repository: https://github.com/weichkai/footPressureSensor

[2] Footwear Sensor - Project SS2025 - https://github.com/natalyss16/footsole